



Facilitator Guide

Sector
Green Jobs

Sub Sector
Other Green Jobs

Occupation
Green Logistics

Reference Id

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Green Logistics Practices

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“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”

Shri Narendra Modi
Prime Minister of India





Certificate

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This Facilitator Guide is meant for understanding green logistics by the trainers in logistics sector and attempt to ensure the availability of all the relevant information to the existing and prospective trainers in green logistics practices. We have compiled the content with inputs from the relevant Subject Matter Experts (SMEs) and industry members to ensure it is the latest and authentic. We express our sincere gratitude to all the SMEs and industry members who have made invaluable contributions to the completion of this Facilitator Guide.

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This Facilitator Guide will help deliver skill-based training with a hope that it will benefit all the stakeholders, such as trainers, and evaluators. We have made all efforts to ensure the publication meets the current quality standards for the successful delivery of QP/NOS-based training programs. We welcome and appreciate any suggestions for future improvements to this Facilitator Guide.



About this Guide

Why Green Logistics Practices?

India has high requirement for a skilled workforce. As per an earlier World Bank study, more than 12 million youth (between 15 years to 29 years) are expected to enter India's workforce every year for the next two decades; of these, only 2.3% receive some formal skills training. The Covid pandemic has also revealed the severe shortage of skilled workforce at the interface of the health, digitalization and **logistics sector**.

Out of these sectors, greening of logistics sector has really picked up pace in the past few years and a lot of companies are making genuine efforts to incorporate the same in their day-to-day operations.

When introducing green skills in any sector there are two parts to it:

1. There are these generic skills which are applicable to all the job roles irrespective of sector such as efficient use of natural resources, efficient use of storage spaces, reducing energy wastage, etc.
2. There are these job roles and sector specific skills that are unique to each job and/ or sector

Hence, this guide focuses on the introduction of generic skills under green skills in the logistics sector.

Green Logistics is the term that describes integrating environmental considerations into the logistics process. The goal of green logistics is to minimize the ecological impact of logistics activities while maximizing efficiency and effectiveness.

Some key benefits of green logistics are as under:

1. Reduction in environmental impact through reduction in greenhouse gas emissions.
2. Green Logistics practices can lead to reduced energy consumption, further reducing emissions and saving money.
3. Improved efficiency through better utilization of resources
4. Green Logistics can help businesses improve their image and reputation.

This guide is designed to incorporate environmentally sustainable practices and principles into logistics education and training. This aims to train existing and future professionals in the field of logistics while emphasizing the importance of reducing environmental impacts, optimizing resources, and promoting sustainable operations within the supply chain. Adopting green skills in logistics sector not only benefits the environment but can also lead to reduced operating costs, improved public perception, and compliance with evolving environmental regulations. It is essential for logistics operators to stay updated on the latest sustainable practices and technologies to remain competitive and environmentally responsible in the logistics and supply chain industry. In that context, this NOS is to introduce generic green skills under logistics sector courses by dovetailing green skill modules to the current and future logistics course curricula.

Symbols Used



**Key Learning
Outcomes**



Steps



Role Play



Tips



Notes



**Unit
Objectives**



Activity



Practical

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Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed



Employability skills workbook





Key Learning Outcomes



At the end of this module, you will be able to:

1. Explain about the concept of Sustainable Development and SDG's.
2. Define Greenhouse Gas Emissions and its sources in Logistics.
3. Identify the environmental impact of Logistics.
4. Explain the concept of Circular Economy and its relevance to Logistics Sector
5. List out the benefits of Circular Economy

UNIT 1.1: Introduction to Sustainable

Unit Objectives

At the end of this unit, you will be able to:

1. Define Sustainable Development and SDG's
2. Explain the roadmap of SDG's from MDG's
3. Discuss how the goals are interconnected in multiple ways (SDG 7 & SDG 13)
4. Brief about the Conference of Parties (COP)

Explain

Tell the learners how this presentation/ course/ qualification will benefit them.

Briefly state your credentials in this area, or explain why learners/candidates should listen to you.

Say

Introduce the module to the learners.

Explain

Emphasize on points especially on punctuality and keeping mobile phones switched off. Ensure that the learners understand the importance of adhering to discipline and code of conduct.

Encourage the learners to ask questions for clear understanding and the proper way of asking without interrupting the class.

Additionally, you may tell the learners to also ensure that they are dressed in proper dress code. You may give examples of good code of conduct. For example, somebody whistling in the classroom is not appropriate.

Ask

You may question the learners if they have any idea about what is Sustainable Development and SDG's.

Explain



Discuss with the Learners about Sustainability, the 3 pillars of sustainability, Sustainable Development, Sustainable Development Goal (SDG's), the 5 pillars of Sustainable Development Goals.

Explain to the learners about how SDG came into existence through MDG's.

Discuss with the learners how SDG 7 and SDG 13 are interconnected.

Notes for Facilitation



Brief the learners about the Conference of Parties (COP).

UNIT 1.2: Overview of Greenhouse Gas Emissions- its sources in Logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Define Greenhouse Gases and GHG Emissions
2. List the cause of rising GHG Emissions
3. Discuss how GHG Emissions causes climate change
4. List the sources of GHG in Logistics
5. Identify how Logistics Sector impacts the environment
6. List the probable areas to adopt for sustainable practices in Logistics Sector

Explain

Discuss what are Greenhouse Gases, Greenhouse effect, Greenhouse Gas Emissions and Global Warming.

You may explain to the learners about what causes rise in emissions and how Greenhouse Gas Emissions causes climate change.

Explain

Discuss about the sources of Greenhouse Gase in Logistics.

You may explain the learners about the environmental impact of Logistics.

Say

Tell the learners about the probable areas to adopt sustainable practices in Logistics Sector.

Notes for Facilitation

Discuss the Indian Scenario on increasing GHG Emissions in Logistics Sector.

UNIT 1.3: Introduction to Renewable Energy & its sources

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the growing significance of renewable energy in today's world
2. Define renewable energy and its sources
3. Explain the difference between the conventional and renewable energy sources.
4. List the benefits of renewable energy.

Say

Tell the learners about the meaning of renewable energy.

Read out and explain to the learners about the sources of renewable energy.

Explain

Briefly Explain the growing significance of renewable energy in today's world.

Describe to the learners about the meaning of conventional energy.

List the difference between conventional energy and renewable energy.

Say

Tell the learners about the key benefits of renewable energy.

Notes for Facilitation

Tell the learners about the India's position in renewable energy

You may show and tell the learners about the renewable installed capacity till Feb, 2024 which may vary year on year as per the targets.

Tell the learners about the inclusion of large hydro power source in the total renewable installed capacity.

UNIT 1.4: Uses of Renewable Energy in Logistics Sector

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the Logistics Sector and Energy needs.
2. Discuss the current energy landscape in Logistics Sector.
3. Explain different renewable energy sources suitable for Logistics.
4. Describe the concept of Circular Economy.
5. Illustrate through an example to calculate electric load of the workplace and identify the solar plant capacity to be installed.

Notes for Facilitation

Provide the learners a brief overview on Logistics Sector and Energy needs.

You may tell the learners about the current energy landscape in Logistics Sector.

Introduce the learners about the key initiatives taken by India in boosting Logistics Sector.

Say

Tell the learners about the different renewable energy sources suitable for Logistics.

Read out and explain to the learners about the benefits of renewable energy in Logistics Sector.

Explain

Briefly explain the concept of Circular Economy

Explain the learners about the application of Circular Economy in Aviation Sector as an example.

Introduce the concept of Circular Economy in Logistics Sector.

Discuss with the learners about how Circular Economy concept are integrated into the Logistics Sector.

Notes for Facilitation



Tell and explain the learners about how different companies are successfully implementing Circular Economy through detailed multiple organisation's case studies.

Explain



Illustrate and explain through an example on how to calculate the electric load of the workplace and identify the solar plant capacity to be installed.

Summarize



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.



2. Introduction to Green Logistics

Unit 2.1 – Brief concept of Green Logistics

Unit 2.2– Overview of Green Supply Chain Management

Unit 2.3– Introduction to Carbon Footprint



Key Learning Outcomes

At the end of this module, you will be able to:

1. Define the basic concept of Green Logistics
2. Discuss about sustainable logistics practices
3. Identify the difference between conventional logistics and green logistics
4. Describe the key initiatives taken for green logistics in India
5. Identify the list of components of green logistics
6. Establish the drivers and key levers for green logistics in Indian context
7. Explain the meaning of green supply chain management, warehouse management, transport/ shipping and their impact on environment
8. Explain about the supply chain performance metrics
9. Establish the components of measuring carbon footprint of a shipping/ transport agency

UNIT 2.1: Brief concept of Green Logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Give an overview of green logistics and sustainable logistics
2. Explain sustainable logistics practices
3. List the difference between conventional logistics and green logistics
4. Explain the key drivers and levers for green logistics in Indian context
5. Discuss the green logistics initiatives taken by India

Explain

Provide a brief overview of green logistics and sustainable logistics to the learners.

List the components of green logistics.

Say

Read out and explain the sustainable logistics practice.

List and discuss the difference between conventional and green logistics to the learners.

Outline and explain the learner about key drivers and levers for green logistics in Indian context.

Notes for Facilitation

Tell the Learners about the green logistics initiatives in India like National Logistics Policy (2022) with its features.

Explain the learners about how these initiatives like NLP- PM Gati Shakti, The Sagarmala project, Bharatmala Yojana will contribute and improve the efficiency and sustainability in Logistics Sector.

Demonstrate

Elaborate the learners about the different Logistics Organisations going green through video.

UNIT 2.2: Overview of Green Supply Chain Management

Unit Objectives

At the end of this unit, you will be able to:

1. Give introduction to green supply chain management
2. Discuss the impact of supply chain management, warehouse management, transport/ shipping on environment
3. Explain the components of green supply chain performance metrics

Explain

Define and explain to the learners about green supply chain management.

Explain

Explain the learners about the environmental impact of supply chain management- which involves emissions through manufacturing, resource depletion, waste generation, water and soil pollution etc.,

warehouse management- emissions through energy consumption, water usage, waste generation, air pollution etc.,

transport/ shipping- emissions in transportation through marine, air, road, rail.

Say

Read out the points given on the slides.

Briefly explain the components of green supply chain performance metrics.

UNIT 2.3 Introduction to Carbon Footprint

Unit Objectives

At the end of this unit, you will be able to:

1. Define carbon footprint
2. Explain the components of measuring carbon footprint of a shipping/ transport agency

Explain

Define and explain Carbon Footprint.

Tell the learners about the two variables of carbon footprint based on which calculation is carried out.

Explain the learners through an example about what activities need to be considered while calculating carbon footprint.

Notes for Facilitation

Tell the learners that World Resource Institute (WRI) and World Business Council for Sustainable Development (WBCSD) created GHG Protocol as an international standard for corporate accounting and reporting emissions.

The Greenhouse Gas Protocol (GHG Protocol) is a globally acknowledged standard for measuring and managing greenhouse gas emissions.

It provides the accounting framework for nearly every GHG standard and program in the world - from the International Standards Organization to The Climate Registry - as well as hundreds of GHG inventories prepared by individual companies as well as GHG Protocol offers an internationally accepted management tool to help businesses to compete in the global marketplace and governments to make informed decisions about climate change.

Say

Tell the learners about the 3 scopes of GHG Protocol through diagram/picture.

Explain



Discuss with the learners about the components to be considered while measuring the carbon footprint of shipping/ transport agency.

Notes for Facilitation



Introduce to the learners about the UNFCCC GHG emissions calculator for calculating the workplace's annual GHG inventory emissions and removals of direct GHGs.

Summarize

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.





3. Steps required for greening the Logistics Sector

Unit 3.1 - Best practices and strategies to make supply chain management green

Unit 3.2 - Implementation of practices under energy/ electricity/ material/ water conservation

Unit 3.3 – The use of renewable energy in the logistics sector

Unit 3.4 - Introduction to circular economy in different areas

Unit 3.5 - Implementation of efficient green logistics



Key Learning Outcomes



At the end of this module, you will be able to:

1. Discuss the green practices and Strategies to Green the Supply Chain Management
2. Identify Sustainable materials in the sector
3. Identify how to Streamline inventory management for a greener supply chain
4. Discuss about Resource Conservation (e.g., material, energy, water usage)
5. Identify how to Implement Energy/ Electricity/material/water Conservation Practices
6. Identify the use of Renewable Energy / solar energy in the sector
7. Identify how to manage different types of Waste effectively
8. Explain how to review internal operations with the point of energy, material and waste management
9. Describe the concept of circular economy and where it can be introduced
10. Establish the concept of green logistics and how efficiently it can be implemented
11. Explain the significance of optimizing logistics operations for sustainability

UNIT 3.1: Best practices and strategies to make supply chain management green

Unit Objectives

At the end of this unit, you will be able to:

1. Introduce green supply chain management
2. Discuss best practices to make supply chain green
3. Discuss strategies to make supply chain green
4. Identify the sustainable materials in the sector
5. Explain how to streamline inventory management for greener supply chain
6. List the suggestions to make work place operations/ activities environment friendly

Explain

Discuss the term green supply chain management and its benefits.

Explain the best practices and strategies for making supply chain green.

Do

Show and explain to the learners through the video about sustainable practices used at workplace.

Say

List down and explain the sustainable materials in the sector

Explain

Identify and discuss with the learners about how to streamline inventory management for green supply chain.

Explain the suggestion to the learners on how to make workplace operations/ activities environment friendly.

UNIT 3.2: Implementation of practices under energy/ electricity/ material/ water conservation

Unit Objectives

At the end of this unit, you will be able to:

1. Define the meaning of resource conservation (e.g., material, energy, water usage)
2. Explain the Implementation of energy/ electricity/ material/ water conservation practices
3. Explain the management of different types of waste
4. Discuss the steps in reviewing internal operations with the point of energy, material and waste management

Explain

Describe resource conservation and its components (in material, energy, waste usage).

Explain the different resource categories with their importance.

Tell the learners how to implement best conservation practices in energy conservation, electricity conservation, material conservation and water conservation.

Say

Read and explain to the learners about the list of green practices to be followed at workplace.

Explain

Talk about the problem of waste - Our work activities generate a lot of waste material. It could be part of the production stage or later when we consume goods as well.

Now offices, factories, public places and even our homes are expected to separate this waste into different types, but its best to avoid the solid waste in the first place.

Solid waste refers to the range of garbage materials—arising from animal and human activities—that are discarded as unwanted and useless. Solid waste is generated from industrial, residential, and commercial activities in a given area, and may be

handled in a variety of ways. As such, landfills are typically classified as sanitary, municipal, construction and demolition, or industrial waste sites.

Waste can be categorized based on material, such as plastic, paper, glass, metal, and organic waste. Categorization may also be based on hazard potential, including radioactive, flammable, infectious, toxic, or non-toxic wastes. Categories may also pertain to the origin of the waste, whether industrial, domestic, commercial, institutional, or construction and demolition.

Explain



Separating wastes will help us remake goods from it, minimise waste, save money and reduce the size of our landfills. Each type of waste has its own process and rules. Please ensure you follow the process as required by your office. This segregated waste is put into different containers and these are colour coded. In other sections you will learn how to do this.

Waste segregation and management is also required by the government and if an office is found to be in violation, then it is a serious issue with fines and more.

Talk about the elements of a Waste Management system – There are 6 important elements of a waste management system. They are:

Waste generation – These are activities involved in identifying materials that are no longer usable and are either gathered for systematic disposal or thrown away.

Onsite handling, storage and processing – These can be activities at the point of waste generation, which facilitate easier collection. For example, waste bins are placed at sites that generate sufficient waste

Waste collection – An important phase of waste management, this includes activities such as placing waste collection bins, collecting waste from those bins, and accumulating garbage in the location where the collection vehicles are emptied

Waste transfer and transport – These are activities that enable the transfer of waste to remote or regional waste Centres in large waste transport vehicles. This will not be in an office or commercial environment.

Waste processing and recovery – These are activities to recover reusable or recyclable materials from the waste stream but are not in the office or commercial environment.

Disposal – This is disposal of waste materials that cannot be recycled in locations such as landfills or waste-to-energy facilities. This will not be in an office or commercial environment.

Explain

Talk about Dry Waste - Dry waste is non-biodegradable and can be typically defined as the materials or waste that does not rot or decay over time. This waste contains very little or almost no moisture content. All the dry waste is recyclable. Some examples are

- Plastic - This could include plastic bags, used cups and plates, Juice or Pet bottles
- Paper - Newspapers, Printed papers, stationary, cardboard boxes
- Metal - This could be metal packs, offcuts and containers
- Foam - Thermocoal and related similar
- Production waste in industries
- Wood and Glass - Boxes, Containers or utensils

Talk about Organic Waste - Organic waste is simply any material that comes from a plant or animal and is biodegradable. Anything that you can put in a compost pile is organic waste. The following can be recycled to compost.

- Food - This is typically food, vegetables and can be sent for composting. Leftover food could also be used for generating biogas.
- Horticulture waste - In buildings or factories, material like grass trimmings, plants, leaves etc., and even bouquet flowers.

Talk about Sanitary and Biowaste - This is hazardous waste and has to be clearly identified and there is special handling process to be carried out for this.

- Sanitary - Diapers, Hair, used tissue from toilets
- Used material cloth, badly soiled paper, waste from cleaning operations
- Biomedical waste from medical facilities

Talk about Agri waste - Proper collection and disposal to avoid burning - Agriculture waste management is an important issue. With the right kind of practices, we can avoid issues like crop burning and also save energy. Some simple steps and a disposal plan can go a long way

- Do not burn garbage or farm waste. It should always be recycled or reused or disposed for landfill.
- Safely remove surface materials like pvc pipes etc., and separately pack them
- Wood – Identify for Salvaging, Chipping or Excavation
- Pruning can be a value resource and identify the possible uses
- Nursery Waste - Plant waste is a valuable resource when composted
- Agricultural plastic can be recycled. Separate them and identify local programme details for this. Livestock feed plastics have specific bags as well. Similarly, greenhouse and nursery film plastics, when clean, can be made into pellets or smaller pieces and other uses.
- Agricultural chemical containers are recycled differently from other agricultural waste and would follow the chemical waste management process.

Talk about E-waste - The term "e-waste" is an short form for "electronic and electrical waste". It is considered waste as the item has no further use and is rejected as useless or excess to the owner in its current condition. E-waste includes almost any household or business item containing circuitry or electrical components with either power or battery supply. These are products that are used in electronic devices and need a special recycling process. E-waste covers six waste categories and each has their own process for recycling.

- Temperature exchange equipment - cooling / heating / freezing equipment like refrigerators, freezers, air conditioners, heat pumps Screens, Monitors - televisions, monitors, laptops, notebooks, and tablets
- Lamps and lights - fluorescent lamps, high intensity discharge lamps / bulbs, and LED lamps
- Large equipment - washing machines, clothes dryers, dish-washing machines, electric stoves, large printing machines, copying equipment, and solar or photovoltaic panels

- Small equipment - vacuum cleaners, microwaves, ventilation equipment, toasters, electric kettles, electric shavers, scales, calculators, radio sets, video cameras, electrical and electronic toys, small electrical and electronic tools, small medical devices, small monitoring and control instruments
- Small IT and telecommunication equipment - mobile phones, Global Positioning Systems (GPS), pocket calculators, routers, personal computers, printers, telephones

Talk about Chemical Waste - Chemical waste is any type of waste that is composed of harmful chemicals and solvents that are the byproducts of large-scale laboratories and manufacturing plants. However, certain refrigerants, batteries and cleaning products qualify as chemical waste, too. This has a special treatment process for each type.

- Spent Solvents
- Chemicals
- Wasted Oils
- Filters for Air and Oil
- ETP (Effluent Treatment Plant) Sludge
- Resins and Paints

Explain

Talk about the importance of Waste Segregation - Waste Segregation is the first step in a compliant waste management plan that will help the environment and save the business money.

Offices and factories have various kinds of containers placed in common areas.

Everyone is encouraged to follow these rules strictly. Each location will also have stickers informing every one of the correct uses. Typically, these could be the following three

- **Talk about Blue Bins** for Dry Waste - The blue-coloured bin is used for segregating dry or recyclable left overs. The Dry recyclable waste includes paper, plastic, cardboard, bubble-wrap etc. The bin would typically be 70L in size.
- **Talk about Green Bins** for Organic Waste - The green-coloured bin is used to dump biodegradable waste. The size of bin will depend on whether employees eat their lunch within the office space, or go out to a canteen or outside.
- **Talk about Red Bins** for Sanitary Waste - Bathrooms in the office premises will have a bin for biomedical waste.
- **Talk about Grey Bin** - E-waste will have dedicated containers marked in grey and bigger in size depending on the type of e-waste devices in the location. The place this bin is kept could be central in the complex. This would enable everyone to make use of this bin.
- **Talk about Yellow Bin / Zone** - Hazardous waste will typically be handled in very specialised ways

Say

Talk about Red Bins and Non-Recyclable Waste - Some materials are not readily recyclable and are appropriate for disposal today. But with research and advancements in technology they could become resources tomorrow.

- For now, the Red Bin material will need to go to the sanitary landfill and made safe.
- At an office or factory site, the red bin containers are stored in a designated area. This will be cleared by the service provider at a designated time and periodicity. This is dependent on the amount of waste generated per day.

Say

- **Talk about Hazardous chemical waste** – It is defined as any substance that poses a health threat to humans or could cause damage to the environment, if not managed correctly. Chemical waste can be generated by many sources, ranging from chemical laboratories and industrial manufacturing to the disposal of household batteries or cleaning fluids. If this waste is allowed to go into our water streams it can impact both the human life and also the water animals. Some things to keep in mind:
- **Reduce the use of hazardous chemicals** - make sure you only use the minimum amounts required, re-use as many chemicals as possible, recycle all recoverable materials, and then dispose of any remaining hazardous substances.
- **Maintain record of all chemicals** - Inventory limits will always be set according to emergency response capacity. Maintain records in software to manage chemical stocks. This ensures the right amount of material is used.
- **Preventing spills and maximizing material usage** - Efficient production scheduling, to ensure chemicals are only taken from storage when they're going to be used, will also help mitigate the risk of leaks and spills. Minimizing the need to transfer materials into different containers, and using pre-weighed packages to reduce handling requirements, will also help reduce the risk.

Talk about Chemical Waste Recycling - When chemicals are recycled, the useful materials that can't be directly re-used, and would otherwise be discarded, are recovered, collected, and separated from the chemical waste, significantly reducing the amount of waste that needs to be treated and discarded. Hazardous waste recycling can be carried out on-site, with the recycled materials being returned to a manufacturing process within the same facility, or waste can be collected and the recycling carried out off-site, by a licensed third-party.

Talk about Chemical waste disposal - While lots of chemical waste can be safely and effectively recycled, other chemical waste will need to be treated and disposed of correctly.

- Hazardous chemical waste treatment is carried out at dedicated Treatment Storage and Disposal Facilities (TSDFs).
- Prevent chemical waste from accumulating at the premises, take regular advantage of local hazardous waste collections; and by properly segregating your waste in your on-site chemical waste storage area, you can remove the risk of cross-contamination.

Talk about the importance of Proper Labelling of Hazardous Waste - must always be correctly labelled:

- Containers must be clearly labeled as hazardous waste.
- The date accumulation begins.
- Each container must be labeled as follows
 - composition and physical state of the waste
 - statement or statements that call attention to the particular hazardous properties of the waste (e.g. flammable, reactive, corrosive)

Explain



List and explain the steps required to review the internal operations in energy, material and waste management

UNIT 3.3: The use of renewable energy/ solar energy in the sector

Unit Objectives

At the end of this unit, you will be able to:

1. Give an introduction to renewable energy in supply chain management
2. Explain the use of renewable energy/ solar energy in supply chain management

Explain

Introduce the renewable energy in supply chain management to the learners.

Explain

Explain the uses of renewable energy/ solar energy in supply chain management with respect to manufacturing & production, transportation & logistics, warehousing & distribution, procurement & sourcing and inventory management.

Do

Show and explain an example to the learners through the video about companies (like NIKE) using sustainable practices used at workplace.

UNIT 3.4: Introduction to circular economy in different areas

Unit Objectives

At the end of this unit, you will be able to:

1. Give an overview of circular economy
2. Establish the areas where the concept of circular economy can be introduced

Notes for Facilitation

Tell the learners that “circular economy” is an industrial system that is restorative or regenerative by intention and design. It replaces the ‘end-of-life’ concept with restoration, shifts towards the use of renewable energy, eliminates the use of toxic chemicals, which impair reuse, and aims for the elimination of waste through the superior design of materials, products, systems, and, within this, business models

Explain

Discuss the different circular strategies in production chain with different R's.

Explain to the learners about the relationship types between circular economy and sustainability.

Say

Read and tell the learners about the areas where the concept of circular economy can be introduced.

UNIT 3.5: Implementation of efficient green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss how to implement an efficient green logistics
2. Give successful examples of companies taking initiatives to make logistics activities green
3. Explain the significance of optimizing logistics operations for sustainability

Explain

Explain the learners about the stages involved in implementing green logistics in an organisation.

Describe various ways to implement sustainability into logistics services.

List the areas of green logistics and its impact

Say

Tell the learners about the benefits of implementing green logistics.

Explain

Discuss the examples of companies taking initiatives to make logistics activities green and their impact.

Explain the significance of optimizing logistics operations for sustainability.

Summarize

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.





4. Identifying green supply and demand networks

Unit 4.1 – Concept of green purchasing in supply chain management

Unit 4.2 – Selection of green logistics partner and transporter



Key Learning Outcomes



At the end of this module, you will be able to:

1. Explain the meaning of green purchasing and its importance
2. Identify the ways for Green Purchasing in Supply Chain Management
3. Identify the best Practices for Green Purchasing
4. Identify the criteria for selecting a supplier with green attributes
5. Explain how to evaluate and select supplier based on green approach in Supply Chain Management
6. Explain how to choose a green logistics partner
7. Identify how to select transporter which are following green practices
8. Describe the map of entire supply chain, identifying suppliers, transportation routes, manufacturing processes, and end consumers.
9. Discuss how to prepare an assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing

UNIT 4.1: Concept of green purchasing in supply chain management

Unit Objectives

At the end of this unit, you will be able to:

1. Define the meaning of green purchasing and its importance
2. Explain the ways for Green Purchasing in Supply Chain Management
3. Discuss the best Practices for Green Purchasing

Explain

Explain to the learners about the concept of green purchasing with its principles and importance.

Discuss the different ways for green purchasing/ procurement in supply chain management.

Introduce to the learners about the best practices to be adopted while purchasing/ procuring goods & services.

UNIT 4.2: Selection of green logistics partner and transporter

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the criteria for selecting a supplier with green attributes
2. Identify the supplier evaluation and selection based on green approach in supply chain management
3. Identify how to select green logistics partner
4. Identify how to select transporter following green practices
5. Discuss the assessment criterion to evaluate suppliers based on their sustainability practices, covering aspects like energy efficiency, waste management, and ethical sourcing.
6. Give an overview of supply chain map
7. Illustrate through Companies using upgraded technology
8. Explain an Implementation of smart logistics solutions
9. Give examples of emerging future technologies in green logistics

Notes for Facilitation

Tell the learners about the different criteria to be adopted for selecting a supplier with green attributes, these are:

General Criteria- Product price, Quality, Reserved Production Capacity

Criteria related to CSR- Discrimination, Child labour and rights, Internal Training Program

Criterial related to environment- Material selection, Design for process, Design for disassembly, Design for recycling and reuse, Packaging.

Criteria related to the production phase of product life cycle- GHG emissions, air pollution, waterborne pollution, waste, energy consumption

Criteria related to the end-of-life phase in product life cycle- Disposal, recycle rate, reuse rate.

Explain

Explain the learners about the process to evaluate and select supplier based on green approach in supply chain management.

Explain to the learners about what factor needs to be considered while selecting green logistics partner such as:

Environmental conditions, social responsibility, Governance and Compliance, Driving innovations, long term roadmap.

Say

Tell the learners about the different companies known for following green logistics practices as an example

Explain

Explain the learners about different factors to be considered in selecting a transporter following green practices

Tell the learners that Intermodal transportation means transporting one set of goods in a steel container using two or more modes of transportation.

Discuss the assessment criteria to be adopted to evaluate sustainable suppliers.

Give an overview of supply chain map i.e., A supply chain map is a visual representation of the entire supply chain, from raw materials to the final product.

Say

Read and explain the components involved in a supply chain map such as: suppliers, manufacturers, distributors and customers

List down and explain the steps to create supply chain map.

Show and explain the learners with a diagram of supply chain mapping for a food and beverage company.

Explain



Discuss with the learners about the environmental impact of supply chain at each stage with examples, these are:

Raw material sourcing, Manufacturing, Distribution and transportation, Warehousing and storage, Retail and customer use, End of life management.

Explain



Explain to the learners what are the emerging key technologies revolutionizing logistics.

Say



Tell the learners a few examples of logistics companies which have adopted upgraded technologies to improve efficiency, reduce costs, and enhance overall operations with its benefits.

Explain



Explain to the learners that smart logistics is the process of adopting new technologies in all operations carried out inside and outside the warehouse, from the receipt of goods at the loading docks to their distribution and delivery to the end customer as per the agreed upon conditions.

Explain the learners about implementing smart logistics solutions through diagram.

Say



Tell the learners about key benefits of smart logistics.

Provide some real-world examples of companies that have implemented intelligent/ smart logistics solutions

Do



Elaborate an example of emerging future technologies in green logistics.

Summarize



Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.





5. Health & safety in green logistics

Unit 5.1 – Occupational health & safety standards in green logistics

Unit 5.2 – Emergency situations in green logistics

Unit 5.3 – Administering first aid

Unit 5.4 – Work Safety procedures and instructions in green logistics

Unit 5.5 – Usage of PPE in green logistics

Unit 5.6 – Housekeeping practices



Key Learning Outcomes



At the end of this module, you will be able to:

1. Identify occupational health & safety standards and requirements for safe work area in green logistics
2. Identify the different emergency situations in green logistics
3. Explain about the fire protection in green logistics
4. Illustrate how to administer first aid
5. Describe the work safety procedures in green logistics
6. Identify the usage of personal protective equipment (PPE) in green logistics
7. Establish the housekeeping practices and how to prevent infection at the workplace
8. Discuss the case study on practice of safety in green logistics

UNIT 5.1: Occupational health & safety standards in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Give an introduction to occupational health & safety standards in green logistics
2. Explain regulations & requirement for safe work area in green logistics

Say

Introduce the module name to the learners and the relevance of maintaining health and safety standards at the project site.

Notes for Facilitation

Tell them the importance of keeping workplace safe and clean.

Inform them about how to communicate the issues and problems to the higher authorities.

Explain

Explain the use of safety procedures, accident and damages.

Ask

Ask them to protect themselves of such hazards by wearing PPE, when required

Notes for Facilitation

Discuss some measures with them they need to take in case of any mishappening.

Demonstrate

Demonstrate the correct method of giving first-aid. Tell them the importance of giving first-aid before the arrival of medical help.

UNIT 5.2: Emergency situations in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss handling of multiple emergencies like fire emergency, security emergency and medical emergency.

Say

Introduce the general steps to be taken in order to be able to handle any emergency at the workplace.

Notes for Facilitation

Tell them the importance of keeping a list of emergency contact numbers

Inform them that the list of emergency contact details should be posted at multiple locations around the workplace.

Demonstrate

Demonstrate a typical emergency procedure.

Tell them that all emergency must be fully detailed in the operation log book.

Demonstrate

Demonstrate the procedure of operating fire extinguisher to the learners.

Tell them that it is important to keep yourself safe while using fire extinguisher.

Explain

Explain to the learners that theft or vandalism of the equipment & tools from the workplace should be dealt as a security threat.

Say

Introduce the learners about the following steps in such event:

- Do not touch or move anything at the scene
- If you are not certain that the intruder has left the scene, leave.
- If people are around at the workplace, raise an alarm
- Notify the police
- Report:
 - Location
 - Nature of the event
 - Your name and the number you are calling from
- Request immediate assistance
- Take action: Follow the instructions of the police personnel. If you are witnessing the crime, move to a safe location and call the police. Provided it is safe to do so, take a picture or record a video

Explain

Explain to the learners about a sudden onset of medical conditions characterized by acute symptoms of sufficient severity such that the absence of medical attention could reasonably be expected to result in:

- placing the patient's health in serious jeopardy,
- serious impairment to bodily functions or serious dysfunction or any bodily organ or part.

Demonstrate

Demonstrate the steps to be followed while administering first aid during medical emergency.

Notes for Facilitation

Inform the learners about the different emergency contact details/ communication in danger for any medical emergency.

UNIT 5.3: Administering first aid

Unit Objectives

At the end of this unit, you will be able to:

1. Explain how to follow the safety precautions at the workplace.
2. Illustrate the First-aid procedure.

Say

Introduce the learners that in a logistics warehouse, any bio-hazardous material can affect your health. Hence, fall and accidents at the workplace are things to be watched out for.

Introduce some common sickness experienced by people working at warehouse are:

- Stomach infection
- Diarrhoea
- Jaundice
- Typhoid
- Worm infestation
- Falls
- Cuts
- Burns
- Bleeding
- Fractures
- Electric shocks

Demonstrate

Demonstrate the procedures that can help in avoiding serious health hazards while working at the workplace.

Say

Tell the learners that preparing a first aid kit is the first step towards being prepared to give first aid.

Inform the learners about the number of items to be present in a first aid kit.

Notes for Facilitation

Tell the learners the following:

- First aid kit should be kept clean, cool and at a dry place.
- First aid kit should be at easily accessible location and everyone in the plant should know it.
- Some items, such as solutions, may have use-by dates.
- First aid kit should be checked regularly and should be replaced when necessary.
- If an item is used from the first aid kit, promptly replace

Demonstrate

Demonstrate the procedures with different illness such as diarrhoea, head injury, heavy bleeding, heatstroke, burns, electric shock

Demonstrate to the learners through the pictures about the methods of how to give CPR if a person is not breathing.

Demonstrate to the learners about the preventive measures and methods to give first aid as per the American Red Cross “five-and-five” approach if a person is choking/ heat stroke, injured with minor cuts, heavy bleeding, abdominal wound, head injury, burns, poisoning, catches snake bite, fractures, electric shock etc.

UNIT 5.4: Work safety procedures and instructions in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. Illustrate work safety procedures and measures

Say

Introduce the concept of 'hazard' to them. Tell them the examples of each types of hazard.

Personal safety hazards are caused due to humans' negligence.

Electrical hazards are caused due to electrical issues such as short circuit, voltage fluctuations, damaged machinery etc.

Non-electrical hazards are other hazards which are not caused due to any electronic component.

Environmental hazards are caused because of environmental factors such as pollution, gas leak etc.

Notes for Facilitation

Inform the learners that a confined space is defined as any place at workplace that is enclosed and has limited access, such as tanks, dry wells, pump rooms

Confined spaces are potentially hazardous as the breathable atmosphere may become compromised, either by depletion of oxygen or the presence of chemical gasses, such as chlorine or hydrogen sulphide.

Explain

Explain the measures to be taken in storing chemical and material.

Discuss about how to manage movement of vehicles within the warehouse.

Illustrate the safety measures of different types of hazards like natural hazards, workplace hazards etc.

Demonstrate

Demonstrate the procedure of operating a fire extinguisher to them.

Tell them that it is important to keep yourself safe while using fire extinguisher.

UNIT 5.5: PPE usage in green logistics

Unit Objectives

At the end of this unit, you will be able to:

1. List the personal protective equipment to be used by a technician.
2. Illustrate the use and putting on personal protective equipment.

Resources to be Used

Show them the constituents of PPE.

Tell them to wear PPE only when required and replace the damaged one.

Explain

Explain the use of every equipment to them such as:

- Helmet will help them to keep their head safe from an unexpected fall.
- Shoes will help them to prevent any slippage.
- Any ring should not be worn on a finger to prevent getting stuck with the object.
- Wear goggles in case of wiring work.
- Hand and arm protection is needed to keep one safe from fire spark.

Notes for Facilitation

Body harness is needed when a solar technician is working on a height and needs protection from falling.

Safety jacket is worn so that one can remain visible from a distance.

UNIT 5.6: Housekeeping practices

Unit Objectives

At the end of this unit, you will be able to:

1. Explain housekeeping and prevention practices
2. Illustrate safety success stories in green logistics

Notes for Facilitation

Housekeeping in simple words means maintaining a house on a daily or long-term basis or looking after its cleanliness, tidiness, upkeep and smooth running.

Tell the learners about different kinds of accidents commonly caused by poor housekeeping:

- Tripping over loose objects on floors, stairs and platforms.
- Articles dropping from above

Explain

Explain different good housekeeping checklist to the learners for a good hygiene at the workplace including a systematic waste disposal.

Explain

Brief and explain to the learners about the safety success story in green logistics through a case study.

Summarize

Summarize the topics taught in the module and show the videos on performing health & safety in green logistics. You may ask the learners to recall what all has been taught.





Employability skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

[Employability skills workbook](#)



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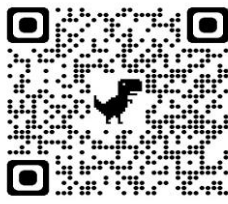


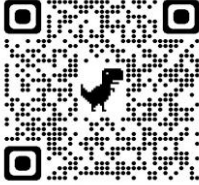
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Annexure



Annexures

Module No.	Unit No.	Topic Name	URL	QR Code
Module 2		Introduction to Green Logistics	https://www.youtube.com/watch?v=12M1r58rVSo	
			https://www.youtube.com/watch?v=f58TVB20A58	
Module 3		Steps required for greening the logistics sector	https://www.youtube.com/watch?v=CfMOMxBfi2g	
			https://www.youtube.com/watch?v=FRfB4ZEnh_A	

Module 4		Identifying green supply and demand networks	https://www.youtube.com/watch?v=pInQN-B9ZaM	
Module 5		Perform Health & Safety in Green Logistics	https://www.youtube.com/watch?v=vWlBgmE6oT8	
			https://www.youtube.com/watch?v=06CqZlj6yxY	
			https://www.youtube.com/watch?v=L1XS5B5D70U	
			https://www.youtube.com/watch?v=1zB7tMH9k_s	
			https://www.youtube.com/watch?v=s0CGgiQOC00	





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